ELECTRODYNAMICS

- To generate a current, some force must be applied on the charges.
- How fast they move in response to the force applied depends on the nature of the material in question.

$$\underbrace{\vec{J}}_{\text{volume current density}} \propto \underbrace{\vec{f}}_{\text{force per unit charge}} \quad \rightsquigarrow \quad \boxed{\vec{J} = \sigma \vec{f}}$$

σ: proportionality factor; depends on the material.
→ It is called "conductivity"

- Most of the time, materials are listed acc. to the reciprocal of σ called resistivity

$$\boxed{\underbrace{\rho}_{\text{resistivity}} = 1/\underbrace{\sigma}_{\text{conductivity}}}$$

Notation: σ and ρ should not be confused with the surface and volume charge densities.

Unit analysis: $\vec{J} \leftrightarrow \dfrac{C}{s\,m^2}$ $\quad \vec{f} \leftrightarrow \dfrac{N}{C} = \dfrac{kg\,m}{C\,s^2}$

$$\sigma \leftrightarrow \frac{C}{\cancel{s}\,m^2}\,\frac{C\,s^{\cancel{2}}}{kg\,m} = \frac{C^2 s}{kg\,m^3} \qquad \rho \leftrightarrow \frac{kg\,m^3}{C^2 s}$$

	Material	ρ
	Copper	1.68×10^{-8}
	Gold	2.21×10^{-8}
Insulators	Wood	$10^{8} - 10^{11}$

Even the insulators have some conductivity. Therefore there is a huge gap between the conductivity of insulators and conductors. Therefore, that's why most of the time metals, for example, can be regarded as perfect conductors. $\sigma \to \infty$.

- The force driving the charges to produce current could be anything. But we will consider electromagnetic force that does the job

$$\vec{F} = Q(\vec{E} + \vec{v} \times \vec{B}) \quad \rightsquigarrow \quad \boxed{\vec{J} = \sigma(\vec{E} + \vec{v} \times \vec{B})}$$

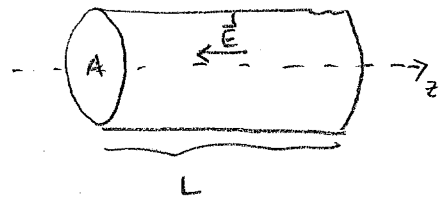
- Most of the time, the velocity of the charges is sufficiently small. (Sometimes this cont. can be significant like in plasmas)

$$\boxed{\vec{J} = \sigma \vec{E}} \quad \text{Ohm's Law}$$

★ Therefore, Ohm's Law is a special case of $\vec{J} = \sigma \vec{f}$.

★ We discussed before $\vec{E} = 0$ inside a conductor. But it was applicable for stationary charges $\vec{J} = 0$. Now, we are considering moving charges. However for perfect conductors $\vec{E} = \vec{J}/\sigma = 0$ even if the current is flowing. In practice, metals are very good conductors that the $\vec{E}$ required to drive current in them is negligible.

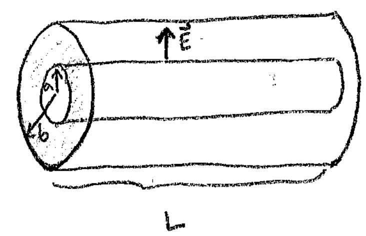
Ex 7.1) A cylindrical resistor of cross-sectional area A and length L made from a material with conductivity σ. If the potential is constant over each end and their difference is V, what current flows? (magnitude of the current)



It is given that $\vec{E}$ is uniform within the wire

$$J = \frac{I}{A} \Rightarrow I = JA = \sigma E A = \sigma \frac{V}{L} A$$

Ex 7.2) Two concentric cylinders are seperated by a material of conductivity σ. If they are kept at a potential difference V, what is the magnitude of the current flowing from one cylinder to the other in a length L? (Assume that the charge per unit length on the inner cylinder is λ.)



Since V is kept at a certain value there will be a certain amount of charge on the both cylinders.

Therefore, the $\vec{E}$ between the cylinders can be calculated as

$$\oint \vec{E}\cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} \rightarrow E\, 2\pi s L = \frac{\lambda L}{\epsilon_0} \Rightarrow \vec{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{s}$$

$$I = \int \vec{J}\cdot d\vec{a} = \sigma \int \vec{E}\cdot d\vec{a} = \frac{\sigma \lambda\, \cancel{2\pi s} L}{\cancel{2\pi}\epsilon_0 \cancel{s}} = \frac{\sigma\lambda}{\epsilon_0} L$$

Btw, potential difference between the cylinders is

$$V(a) - V(b) = V = -\int_b^a \vec{E}\cdot d\vec{l} = -\int_b^a \frac{\lambda}{2\pi\epsilon_0 s} ds = -\frac{\lambda}{2\pi\epsilon_0}\int_b^a \frac{ds}{s} = -\frac{\lambda}{2\pi\epsilon_0}[\ln a - \ln b] = \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{b}{a}\right)$$

If we merge these two equations.

$$\frac{\lambda}{\epsilon_0} = \frac{2\pi V}{\ln(b/a)} \rightarrow I = \frac{2\pi V}{\ln(b/a)}\sigma L = \frac{2\pi\sigma L}{\ln(b/a)} V$$

- As above examples illustrate, the total current from one electrode to the other is proportional to the potential difference between them.

In Ex 7.1: $I = \left(\frac{\sigma A}{L}\right) V$ In Ex 7.2: $I = \left[\frac{2\pi\sigma L}{\ln(b/a)}\right] V$

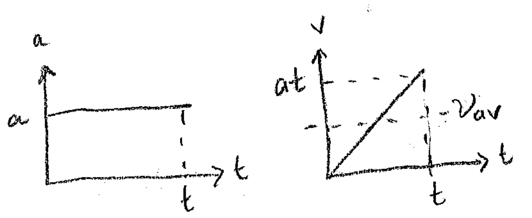
So we can write the proportionality btw. I and V as:

$\boxed{V = IR}$ ⤳ More familiar form of Ohm's law

R: Resistance. is measured in ohms $(\Omega) \equiv \frac{V}{A}$: volt per ampere.

- For steady currents: $\vec{E} = \frac{1}{\sigma}\vec{J} \rightarrow \vec{\nabla}\cdot\vec{E} = \frac{1}{\sigma}\underbrace{\vec{\nabla}\cdot\vec{J}}_{0} \Rightarrow \vec{\nabla}\cdot\vec{E} = 0 \Rightarrow \rho = 0$
- If $\rho = 0$ any unbalanced charge stays on the surface. We proved this for stationary charges ~~but~~ using the fact that $\vec{E} = 0$. It is still true when we allow charges to move
- Ohm's Law implies that constant field produces constant current, which suggests constant velocity. But $\vec{E}$ applies a force $q\vec{E}$ on charges and according to Newton's 2nd law the charge must accelerate. What is happening in here? Is there a contradiction?
- We have to consider the frequent collisions e^- make as they pass down the wire. They are accelerate constantly in between collisions but they are obliged to stop and start accelerating again and again. Therefore, their average speed stays constant.

$$\lambda = \frac{1}{2}at^2 \rightarrow t = \sqrt{\frac{2\lambda}{a}}$$ time passed between 2 collisions.



$$v_{av} = \frac{1}{2}at = \frac{1}{2}a\sqrt{\frac{2\lambda}{a}} = \sqrt{\frac{4\lambda a^2}{\cancel{4}\cancel{a}}} = \sqrt{\frac{\lambda a}{2}}$$

- $v_{av} \propto \sqrt{a} \Rightarrow$ current is proportional to the square root of the field

$$\vec{I} \propto \sqrt{a} \propto \sqrt{E}$$

- But this is not logical because if $I \propto \sqrt{E}$ this creates a huge current with great velocities because what actually happens is e^- are already moving $\not\parallel$ in the wire of their thermal energies. But since v's are random they average to zero. Net DRIFT VELOCITY is a tiny bit. So time between collisions is actually much shorter than we suppose.

$$t = \frac{\lambda}{v_{thermal}} \rightarrow v_{aver} = \frac{1}{2}a\cdot\frac{\lambda}{v_{thermal}} = \frac{a\lambda}{2v_{thermal}}$$

- If there are n molecules per unit volume and f free electrons per molecule each e^- have charge q and mass m

$$\vec{J} = nfq\vec{v}_{ave} = nfq\frac{a\lambda}{2v_{thermal}}\frac{\vec{F}}{m} = \underbrace{\left(\frac{nfq^2\lambda}{2mv_{thermal}}\right)}_{\sigma}\vec{E}$$

- This is not an accurate formula for σ but it has the basic ingredients so that $J \propto n$ and $J \propto \frac{1}{v_{thermal}}$

• As a result of ^(all collisions) the work done by electrical force is converted into heat in the resistor (or circuit).

$$V = \frac{W}{q}$$ work done per unit charge

I: charge flowing per ~~unit time~~.

$$W = qV \Rightarrow \frac{W}{\cancel{s}\,t} = \frac{qV}{\cancel{s}\,t} \Rightarrow \boxed{P = IV = I^2R}$$ → How bulb works

Power delivered.

Joule heating law

I : amperes

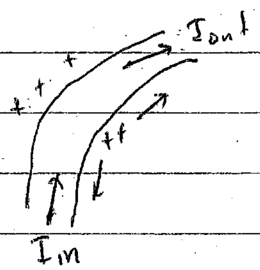
R : ohms

P : Joules per second (1W: watts)

Electromotive Force

[a battery hooked up to a circuit (light bulb)]

• Why the current is the same around a circuit all the time? We hook a battery ~~to~~ a circuit, ~~and~~ if ^(only) the battery has the driving force? Assume that we switched off on the circuit and assume battery started pushing the charges in one direction



If charges pile up at a bend then some $\vec{E}$ would be generated so that this field ~~favors the charges~~ opposes the charges coming in and promotes the ones flowing out

There are two forces involved in driving the current around ~~a~~ circuit

$$\vec{f} = \vec{f_s} + \vec{E}$$

$\vec{E}$ → Electrostatic force (serves to smooth out the flow and communicate the influence of the source to distant parts of the circuit)

$\vec{f_s}$ → source. (confined to a portion of the circuit)

$$\mathcal{E} \equiv \oint \vec{f}\cdot d\vec{l} = \oint \vec{f_s}\, d\vec{l} + \cancel{\oint \vec{E}\cdot d\vec{l}} = \oint \vec{f_s}\cdot d\vec{l}$$

$\mathcal{E}$ → electromotive force (emf). (It isn't a force actually it is the integral of force per unit charge)

Within an ideal source or emf (a resistanceless battery) net force on charges is zero.

$$\vec{J} = \sigma \vec{f}$$

$\sigma \to \infty$

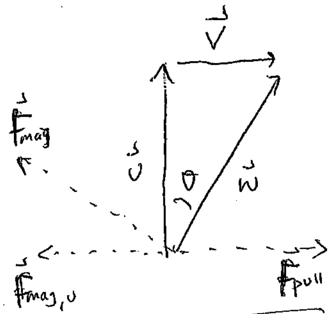
Therefore $0 = \vec{f_s} + \vec{E} \Rightarrow \vec{E} = -\vec{f_s}$

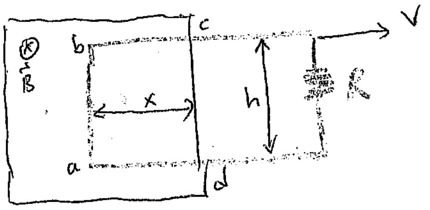
The potential difference :

$$V = -\int_a^b \vec{E}\cdot d\vec{l} = \int_a^b \vec{f_s}\cdot d\vec{l} = \oint \vec{f_s}\cdot d\vec{l} = \mathcal{E}$$

⇒ The function of a battery is to establish and maintain a voltage difference equal to the electromotive force

Electromotive force (motional)





Work done by $\vec{F}_{mag}$

$$W_{mag} = \oint \vec{F}_{mag} \cdot d\vec{\ell} = \underbrace{(qVBh)}_{\text{Lorentz force}}$$

Magnetic force moves the charges inside the circuit. But we know that ~~mag~~ they do no work.

→ The work here is done by the one who is pulling the loop.

Let us show this:

Because of the vertical component of $\vec{w}$ ($\vec{u}$) there will be a force in

$$\left|\vec{F}_{mag,u}\right| = quB$$

* If loop is getting out of the $\vec{B}$ with a constant velocity.

$$\left|\vec{F}_{pull}\right| = \left|\vec{F}_{mag,u}\right|$$

* Therefore, the work done by $\vec{F}_{pull}$

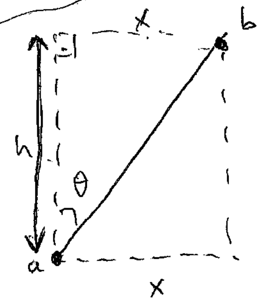
$$W_{pull} = \int \vec{F}_{pull} \cdot d\vec{\ell} = quB\, h \underbrace{\tan\theta}_{\frac{v}{u}}$$

$$= q\cancel{u}Bh\frac{v}{\cancel{u}}$$

$$= qBhV$$

$$\Rightarrow \boxed{W_{pull} = W_{mag}}$$

$\tan\theta = \frac{x}{h}$



We need to write x in terms of h.

* Work done per unit charge inside a circuit is called electromotive force. It does not matter what kind of ~~mechanism~~ driving force did the work.

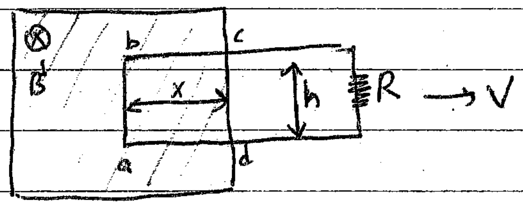
$$\boxed{\mathcal{E} \equiv \oint \vec{f}_s \cdot d\vec{\ell}}$$

* Unit analysis: It is related with potential

• Because it is the line integral of $\vec{f_s}$, $\mathcal{E}$ can be interpreted as the work done per unit charge by the source.

Motional emf

~~It is generated~~ Generators expoit "motional emf" which arise when we move a wire through a magnetic field.



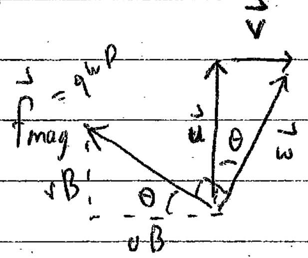
If the entire loop is pulled to right the charges in segment ab will experience a magnetic force whose vertical compon qVB drives current around the loop in the clockwise direction (Let us assume positive charges are creating the current)

• The emf is $\mathcal{E} = \oint \underbrace{f_{mag}}_{\text{force per unit charge}} \cdot d\vec{l} = VBh$

$$\boxed{\frac{N}{C} m = kg \frac{m}{s^2} \frac{m}{C} = \frac{J}{C}}$$

• ~~Lorentz~~ $\vec{B}_{mag}$ is responsible for establishing the emf but we know that magnetic forces do no work. The work here is done by the one who is pulling the loop.

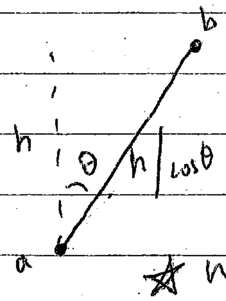
• Charges going up because of the current but they also go right because of $\vec{v}$



To counteract the force quB the person who is pulling the wire must exert a force per unit charge

$$f_{pull} = uB$$

$v = \frac{h}{t}$



Work done by pulling force

$$\int \vec{f}_{pull} \cdot d\vec{l} = uB \frac{h}{\cos\theta} \cos(90-\theta) = uB \frac{h}{\cos\theta} \sin\theta$$

$$= uBh\tan\theta = uBh\frac{v}{u} = BhV = \mathcal{E}$$

☆ Work done per unit charge is exactly equal to the emf ☆

There is a particularly nice way of expressing the emf generated in a moving loop. Let Φ be the flux of $\vec{B}$ through the loop

$$\Phi \equiv \int \vec{B} \cdot d\vec{a} \text{ : magnetic flux}$$

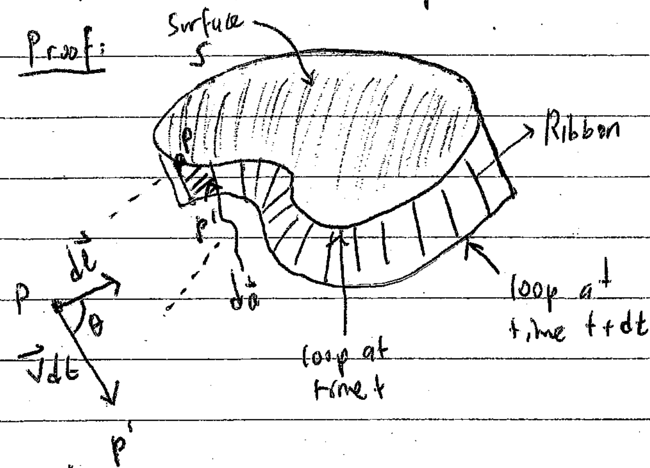
$$\Phi = Bhx \qquad \frac{d\Phi}{dt} = Bh\frac{dx}{dt} = -BhV$$

☆ Emf is the negative of time rate of change of magnetic flux through the loop.

$$\boxed{\mathcal{E} = -\frac{d\Phi}{dt}}$$ "flux rule" for the motional emf

- We ~~proved~~ derived flux rule for a rectangular loop. But it is also applicable for non-rectangular loops moving in arbitrary directions through nonuniform magnetic fields. In fact the loop need not even maintain a fixed shape

Proof:



We will calculate flux at time t by using the surface S and the flux at time $t+dt$ by using the surface + the ribbon

$$d\Phi = \Phi(t+dt) - \Phi(t) = \Phi_{ribbon} = \int_{ribbon} \vec{B}\cdot d\vec{a}$$

$$(\vec{v}dt)\times d\vec{l} = (\vec{v}\times d\vec{l})\,dt = d\vec{a} \Rightarrow d\Phi = \int_{ribbon} \vec{B}\cdot(\vec{v}\times d\vec{l})\,dt$$

$$\Rightarrow \frac{d\Phi}{dt} = \int_{ribbon} \vec{B}\cdot(\vec{v}\times d\vec{l})$$

$\vec{v}$: velocity of the wire

$\vec{u}$: velocity of charge going down the wire

$\Big\}\ \vec{w} = \vec{v}+\vec{u}$: net velocity of the charge.

Since $\vec{w} = \vec{v}+\vec{u}$ and $\vec{u}$ is parallel to $d\vec{l}$

$$\frac{d\Phi}{dt} = \int_{ribbon} \vec{B}\cdot\left[(\vec{w}-\vec{u})\times d\vec{l}\right] = \int_{ribbon} \vec{B}\cdot\left[\vec{w}\times d\vec{l} - \underbrace{\vec{u}\times d\vec{l}}_{0}\right]$$

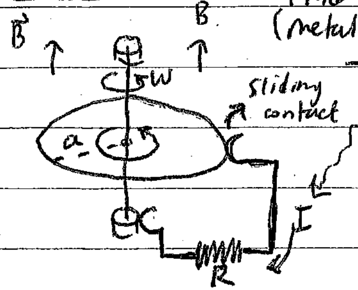
$$= \int_{ribbon} \vec{B}\cdot(\vec{w}\times d\vec{l}) \qquad \text{use the identity } \vec{B}\cdot(\vec{w}\times d\vec{l}) = -(\vec{w}\times\vec{B})\cdot d\vec{l}$$

$$\Rightarrow \frac{d\Phi}{dt} = -\oint_{\substack{\text{along the small} \\ \text{element of} \\ \text{one on the ribbon}}} \underbrace{(\vec{w}\times\vec{B})}_{\text{magnetic force per unit charge}}\cdot d\vec{l} = -\oint \vec{f}_{mag}\cdot d\vec{l} = -\mathcal{E} \quad ✓$$

- To resolve the $-$ sign ambiguity and the ambiguity in the definition of the direction of $d\vec{a}$ use right-hand rule: If our fingers define the positive direction around the loop, our thumb indicates the direction of $d\vec{a}$. If emf come out negative, it means the current will flow in the negative direction around the circuit.

Ex 7.4

- Find the current in the resistor.



(metal disc) → current carrying charges

→ e^-'s in the disc will be acted upon a magnetic force towards the edge of the disc.

$$\mathcal{E} = \int_0^a \vec{f}_{mag}\cdot d\vec{l} = \omega B\int_0^a s\,ds = \frac{\omega B a^2}{2} \Rightarrow I = \frac{\mathcal{E}}{R} = \frac{\omega B a^2}{2R}$$

$$\vec{f}_{mag} = \vec{v}\times\vec{B} = \omega s B\,\hat{s}$$

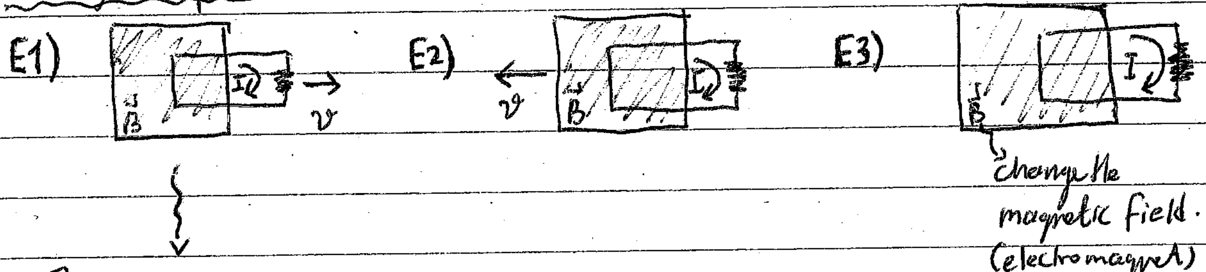
$$\frac{ds}{dt} = \frac{d\phi s}{dt} = \omega s$$

- Keep in mind that flux rule assumes the current flows ~~..~~ along a well-defined path but in this last example current spreads out over the whole disc. And it isn't clear in the last example what the "flux through the circuit" would mean in this context.

Eddy currents : Take a piece of aluminum and ~~....~~ move it in a nonuniform magnetic field. Currents will be generated in the material; and we would feel a kind of drag force as if we were moving it inside a viscous liquid. This happens because of the $\vec{f}_{pull}$ force we discussed above.

Electromagnetic Induction

Faraday's Experiments



This experiment is explained with the flux rule:

$\mathcal{E} = -\frac{d\Phi}{dt}$ and this explanation is also applicable for (E2). ~~All that matters is~~ the relative motion of the magnet and the loop. But Faraday knew nothing about the relativity and maybe this was a nice coincidence because if we move the wire it is obvious that it is the magnetic force setting up the magnetic field. But if the loop is stationary the force cannot be magnetic - stationary charges do not experience magnetic field. So what? what sort of field exerts a force on charges at rest? Electric fields!

→ Faraday came up with the genius idea that

A CHANGING MAGNETIC FIELD INDUCES AN E-FIELD!

→ It is this "induced" field that accounts for the emf in (E2)

→ Faraday found empirically that emf is again equal to the rate of change of flux.

$$\mathcal{E} = \oint \vec{E}.d\vec{l} = -d\Phi/dt$$

- We know that $\Phi \equiv \int \vec{B}.d\vec{a}$

$$\oint \vec{E}.d\vec{l} = -\frac{d}{dt}\int \vec{B}.d\vec{a} = -\int \frac{\partial B}{\partial t}.d\vec{a}$$ → Faraday's Law in integral form.

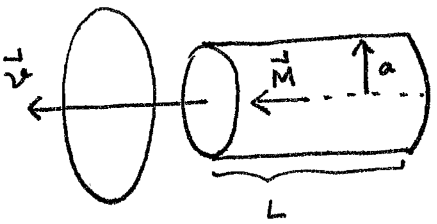
- If we apply Curl theorem: $\int (\vec{\nabla}\times\vec{A}).d\vec{a} = \oint \vec{A}.d\vec{l}$

$\Rightarrow \boxed{\vec{\nabla}\times\vec{E} = -\partial B/\partial t}$ → Faraday's Law in differential form.

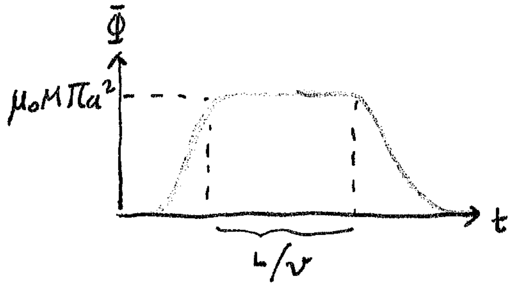
- It reduces to $\oint \vec{E}.d\vec{l} = 0$ (old rule about E-fields) or in diff. form $\vec{\nabla}\times\vec{E} = 0$ in the static case ($\vec{B}$ = constant)
- In (E3) magnetic field changes entirely for different reasons but Faraday's law still applies: An ~~emf will be induced~~ E-field will be induced giving rise to an emf $-d\Phi/dt$.

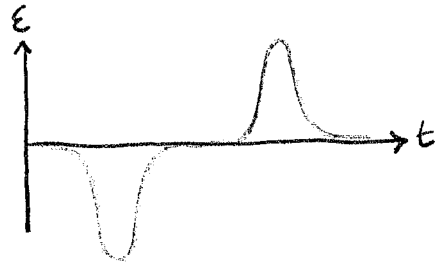
UNIVERSAL FLUX RULE: Whenever (and for whatever reason) the magnetic flux through a loop changes an emf $\mathcal{E} = -\frac{d\Phi}{dt}$ will appear in the loop.

Example 7.5) A long cylindrical magnet of length L and radius a carries a uniform magnetization $\vec{M}$ parallel to its axis. It passes at constant velocity through a circular wire ring of slightly larger diameter. Graph emf induced in the ring, as a function of time.



- Since $\vec{K}_b = M\hat{\phi}$ the magnetic field is the same as that of a long solenoid with surface current $\vec{K}_b = M\hat{\phi}$. So field inside is $\vec{B} = \mu_0\vec{M}$. So flux through the ring is zero when the magnet is far away it builds up to a max of $\mu_0 M \pi a^2$ and it drops back to zero again as the magnet passes away.

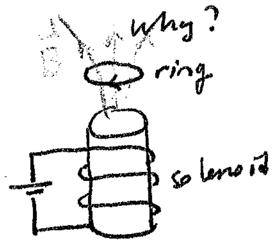




Lenz Law: Nature does not like changes.

$\Rightarrow$ The induced current will flow in such a direction that the flux it produces tends to cancel the change.

Example 7.6) The jumping ring: Think of an iron core and we wind a solenoidal coil around it. (Iron core just magnifies the magnetic field.) If we place a metal ring on top and if we hook the solenoid to a battery the ring will jump in the air



The Induced Field

Faraday's discovery tells us that there are really two distinct kinds of electric fields.

i) Those attributable directly to electric charges $\rightarrow$ Can be calculated with Coulomb's law.

ii) Those associated with changing magnetic field. $\rightarrow$ $\vec{\nabla}\times\vec{E} = -\partial\vec{B}/\partial t$

and $\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

★ Curl alone is not enough to determine a field - we must also specify the divergence.

As long as $\vec{E}$ is a pure Faraday field (it appears because of the change in magnetic flux with $\rho = 0$) Gauss Law says

$$\vec{\nabla}\cdot\vec{E} = 0$$

$\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$	$\vec{\nabla}\times\vec{E} = 0$
$\vec{\nabla}\cdot\vec{B} = 0$	$\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

An also for magnetic fields we know $\vec{\nabla}\cdot\vec{B} = 0$

★ Faraday-induced electric fields are determined by $-\partial\vec{B}/\partial t$ in exactly the same way as magnetostatic fields are determined by $\mu_0\vec{J}$.

- If symmetries in the problems permit us to do that we can use all tricks ~~that~~ associated with Ampere's law in integral form,

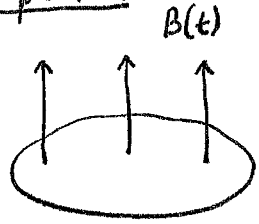
$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$$

but this time it's Faraday's law in integral form

$$\oint \vec{E}\cdot d\vec{l} = -d\Phi/\partial t$$

- The rate of change of (magnetic) flux through the Amperian loop plays the role formerly assigned to $\mu_0 I_{enc}$.

Example 7.7)



A uniform magnetic field $\vec{B}(t)$ fills the shaded circular region. If $\vec{B}(t)$ is changing with time, what is the induced electric field?

$\vec{E}$ points in the circumferential direction, just like magnetic field inside a long straight wire carrying a uniform current density. We simply draw an Amperian loop of radius s and apply Faraday's Law:

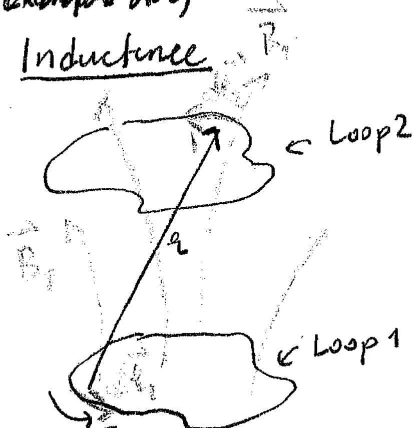
$$\oint \vec{E}\cdot d\vec{l} = E\,2\cancel{\pi} s = -\frac{d\Phi}{dt} = -\frac{d}{dt}\left(\pi s^2 B(r)\right) = -\cancel{\pi} s^{\cancel{2}}\frac{dB}{dt}$$

Therefore $\vec{E} = -\frac{s}{2}\frac{dB}{dt}\hat{\phi}$

If $\vec{B}$ is increasing $\vec{E}$ runs clockwise

~~Example 7.8)~~

Inductance



If we run a current in Loop 1 it produces $\vec{B}_1$.

$$\vec{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\vec{l}_1 \times \hat{r}}{r^2} \Rightarrow B_1 \propto I_1$$

Therefore the flux through Loop 2

$$\Phi_2 = \int \vec{B}_1 \cdot d\vec{a}_2 \Rightarrow \Phi_2 \propto B_1 \propto I_1$$

$$\Rightarrow \Phi_2 \propto I_1 \rightarrow \boxed{\Phi_2 = M_{21} I_1}$$

mutual inductance

$$\Phi_2 = \int \vec{B}_1 \cdot d\vec{a}_2 = \int (\vec{\nabla}\times\vec{A}_1)\cdot d\vec{a}_2$$ by using Curl theorem $\int(\vec{\nabla}\times\vec{A})\cdot d\vec{a} = \oint \vec{A}\cdot d\vec{l}$

$$= \oint \vec{A}_1 \cdot d\vec{l}_2$$

we know that ~~$\vec{A}(r) = \frac{\mu_0}{4\pi}\int \frac{\vec{I}(\vec{r}')}{r} dl'$~~ for line currents $\vec{A}(r) = \frac{\mu_0 I}{4\pi}\int \frac{1}{r} d\vec{l}'$

$$\Rightarrow \vec{A}_1 = \frac{\mu_0 I_1}{4\pi}\oint \frac{d\vec{l}_1}{r} \Rightarrow \Phi_2 = \frac{\mu_0 I_1}{4\pi}\oint\left(\oint \frac{d\vec{l}_1}{r}\right)\cdot d\vec{l}_2 \Rightarrow \boxed{M_{21} = \frac{\mu_0}{4\pi}\oint\oint \frac{d\vec{l}_1 \cdot d\vec{l}_2}{r}}$$

Neumann formula

$$M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\vec{l}_1 \cdot d\vec{l}_2}{\imath}$$

: It involves a double line integral – one integration around loop 1, the other around loop 2.

It is not very useful for application but it reveals two important things:

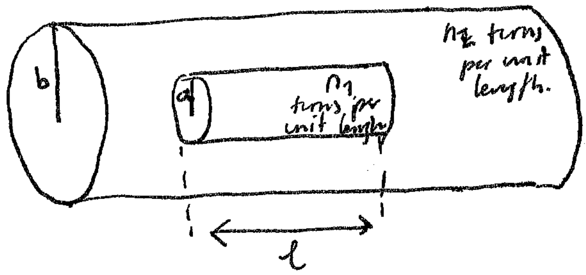
(i) M_{21} is a purely geometrical quantity, having to do with the sizes, shapes and relative positions of 2 loops

(ii) The integral wouldn't change if we switch the roles of loops 1 and 2

$$\Rightarrow \boxed{M_{21} = M_{12}}$$

$\Rightarrow$ whatever the shapes and positions of the loops, the flux through 2 when we run a current I around 1 is identical to the flux through 1 when we send the same current I around 2. ~~we ma~~ Therefore, we can simply drop the subscripts and call them both M.

Ex 7.10)



Current I flows in the short solenoid. What is the flux through the long solenoid.

Inner solenoid is short it has a very complicated field. + it puts a different amount of flux through each turn of the outer solenoid. However, if we exploit the equality of mutual inductances (assume the current flows through the long solenoid)

$B = \mu_0 n_2 I$: field inside the long solenoid it is ~~constant~~ constant

$B \pi a^2 = \mu_0 n_2 I \pi a^2$: flux through a single loop of the short solenoid.

There are $n_1 l$ turns in the short solenoid

$$\Rightarrow \Phi = \mu_0 n_2 I \pi a^2 n_1 l$$

This would be exactly the flux through the long solenoid if there was a current ~~in the~~ of I in the short solenoid.

Therefore the mutual inductance is

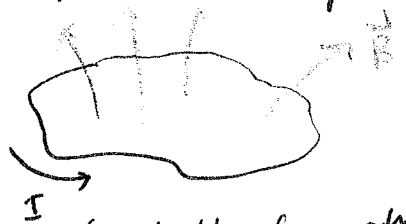
$$M = \cancel{\mu_0 n_2 \pi a} \; \mu_0 n_1 n_2 \pi a^2 l$$

If we change the current in loop 1 the flux through loop 2 will vary accordingly. Faraday's law says that changing flux will induce an emf in loop 2:

$$\boxed{\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M\frac{dI_1}{dt}}$$

=> **<u>RESULT</u>**: Every time we change the flux in loop 1 an ~~I~~ induced current flows in loop 2. even though ~~no~~ there are no wires connecting them.

★ A changing current not only induces an emf in any nearby loops, it also induces an emf in the source loop itself.



- once again the field (and therefore also the flux) is proportional to the current

$$\Phi \propto I \rightarrow \boxed{\Phi = LI}$$

self-inductance (or simply inductance)

- Like M, L also depends on the geometry (size and shape) of the loop.
If the current changes the emf induced in the loop is

$$\mathcal{E} = -\frac{d\Phi}{dt} = -L\frac{dI}{dt} \qquad \boxed{\mathcal{E} = -L\frac{dI}{dt}}$$

- Inductance is measured in henries $(H) \equiv \frac{Vs}{A}$

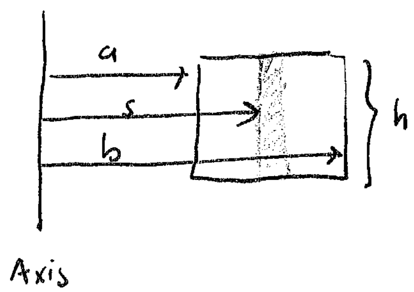
<u>Example 7.11</u> Find the L of a toroidal coil with rectangular cross section (inner radius a outer radius b, height h) which carries a total of N turns.

$$B = \frac{\mu_0 N I}{2\pi s} \quad : \text{field inside the toroid}$$

$$\int \vec{B}\cdot d\vec{a} = \frac{\mu_0 N I}{2\pi} h \int_a^b \frac{1}{s}\, ds = \frac{\mu_0 N I h}{2\pi} \ln\left(\frac{b}{a}\right) \quad : \text{flux through a single turn}$$

The total flux is N times this

$$L = \frac{\Phi}{I} = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$$



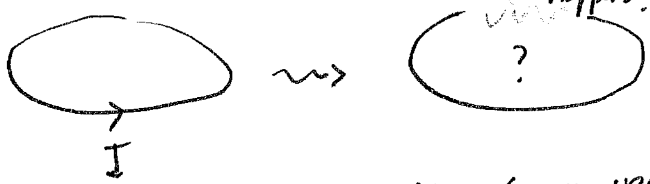
☆ Inductance (like capacitance) is an intrinsically positive quantity.

Lenz Law, which is enforced by the minus sign dictates that the emf is in such a direction as to oppose any change in current. For this reason, it is called a back emf. Whenever we try to alter the current in a wire, we must fight against this back emf.

☆ Inductance plays kind of the same role in electric circuits that mass plays in mechanical systems: The greater L is, the harder it is to change the current, just as the larger the mass, the harder it is to change an object's velocity

Example 7.12)

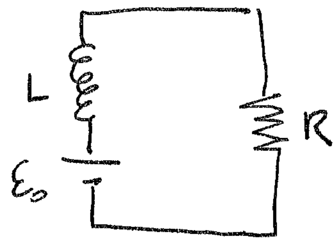
If someone cuts the wire ~~what~~ what happens?



I immediately drops to zero but $\frac{dI}{dt}$ is not zero at the time we cut it. Electromagnetic induction will try to keep the current going even if it has to jump the gap in the circuit!

That's why we usually see a spark when we unplug an ~~electronicd~~ electric device

Oppositely: What happens if we connect a battery (which supplies a constant emf $\mathcal{E}_0$) to the below circuit?



$$V = IR$$

$$\mathcal{E}_0 - L\frac{dI}{dt} = IR \quad \therefore \text{ First order diff eq.}$$

$$\frac{dI}{dt} = \frac{\mathcal{E}_0}{L} - \frac{R}{L}I$$

$$\frac{\mathcal{E}_0}{L} - \frac{R}{L}I = m$$

$$\frac{dm}{dI} = -\frac{R}{L}$$

$$dI = -\frac{L}{R}\,dm$$

$$\frac{dI}{\left(\frac{\mathcal{E}_0}{L} - \frac{R}{L}I\right)} = dt \rightarrow -\frac{L}{R}\frac{dm}{m} = dt$$

$$\frac{dm}{m} = -\frac{R}{L}dt \rightarrow \int\frac{dm}{m} = -\frac{R}{L}\int dt + C$$

$$\ln m = -\frac{R}{L}t + C$$

$$m = e^{-\frac{R}{L}t}\cdot \overbrace{e^{C}}^{k} \rightarrow \frac{\mathcal{E}_0}{L} - \frac{R}{L}I = e^{-\frac{R}{L}t}k$$

$$\frac{R}{L}I(t) = \frac{\mathcal{E}_0}{L} - ke^{-\frac{R}{L}t} \rightarrow I(t) = \frac{\mathcal{E}_0}{R} - \frac{kL}{R}e^{-\frac{R}{L}t}$$

k

$$\frac{kL}{R} = -\frac{\mathcal{E}_0}{R}$$

Energy in Magnetic Fields (See the back page)

$$V = \frac{W}{q} \rightarrow W = qV$$

$$\frac{dW}{dt} = \frac{dq}{dt} V = I \underset{=}{V} \overset{\leftarrow -\mathcal{E}}{} = -\mathcal{E} I = L I \frac{dI}{dt}$$

$$\int \frac{dW}{\cancel{dt}} \cancel{dt} = L \int I \frac{dI}{\cancel{dt}} \cancel{dt}$$

integrate over the time after we start with ~~which~~ a zero current until the current I builds up.

$$\boxed{W = L \frac{I^2}{2}}$$

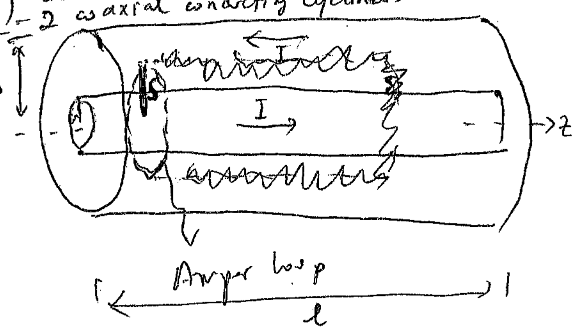
Possible to write this (without proof) pg 317-318

$$\boxed{W_{mag} = \frac{1}{2\mu_0} \int_{\text{all space}} B^2 d\tau} \longleftrightarrow W_{elec} = \frac{\varepsilon_0}{2} \int E^2 d\tau$$

electrostatic counterpart

→ Energy is stored in the magnetic field in the amount $\frac{B^2}{2\mu_0}$ per unit volume

Ex 7.13) – A long coaxial cable. 2 coaxial conducting cylinders –



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc.}$$

$$B \, 2\pi s = \mu_0 I$$ → Find the magnetic energy stored in a section of length l.

$$\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

$$\frac{B^2}{2\mu_0} = \frac{\mu_0^{\cancel{2}} I^2}{4\pi^2 s^2} \frac{1}{2\cancel{\mu_0}} = \frac{\mu_0 I^2}{8\pi^2 s^2}$$; energy per unit volume.

Infinitesimal energy:

$$s \, ds \, d\phi \, dz \, \frac{\mu_0 I^2}{8\pi^2 s^2} \xrightarrow[\text{energy}]{\text{total}} W = \frac{\mu_0 I^2}{8\pi^2} \int_a^b \int_0^{2\pi} \int_0^l \frac{\cancel{s}}{s^{\cancel{2}}} ds \, d\phi = \frac{\mu_0 I^2}{\underset{4}{\cancel{8}}\pi^{\cancel{2}}} l \, \cancel{2\pi} \ln\left(\frac{b}{a}\right)$$

$$= \frac{\mu_0 I^2}{4\pi} l \ln\left(\frac{b}{a}\right)$$

we can write L by using above equation:

$$W = L \frac{I^2}{2} \Rightarrow \frac{\mu_0 \cancel{I^2}}{\underset{2}{\cancel{4}}\pi} l \ln\left(\frac{b}{a}\right) = L \frac{\cancel{I^2}}{\cancel{2}} \Rightarrow L = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right)$$

Maxwell's Equations:

The laws we have encountered so far are

(i) $\vec{\nabla}\cdot\vec{E} = \frac{1}{\epsilon_0}\rho$ (Gauss Law)

(ii) $\vec{\nabla}\cdot\vec{B} = 0$

(iii) $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$ (Faraday's law)

(iv) $\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$ (Ampère's Law)

The law's of electrodynamics before Maxwell.

- The basic problem with these laws is:

$$\underbrace{\vec{\nabla}\cdot(\vec{\nabla}\times\vec{E})}_{0} = -\frac{\partial}{\partial t}\underbrace{(\vec{\nabla}\cdot\vec{B})}_{\text{from (ii)}} \Rightarrow \text{Everything is okay}$$

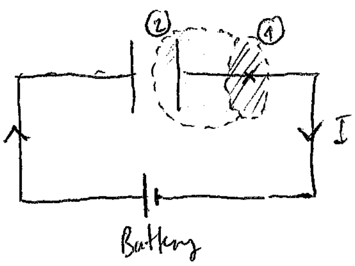
However,

$$\underbrace{\vec{\nabla}\cdot(\vec{\nabla}\times\vec{B})}_{0} = \mu_0\underbrace{(\vec{\nabla}\cdot\vec{J})}_{\text{But } \vec{\nabla}\cdot\vec{J} \text{ is not always zero.}}$$

$$\boxed{\vec{\nabla}\cdot\vec{J} = -\partial\rho/\partial t} \quad \text{continuity equation}$$

→ For steady currents $\vec{\nabla}\cdot\vec{J}=0$ but if we go beyond magnetostatics there must be a problem with Ampère's Law!

Another way to see why Ampère's Law fails for nonsteady currents, we can consider following example:



Try to apply $\oint\vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$ for surfaces ① and ②.

For ①: There is no problem. → $I_{enc} = I$

For ②: Remember that

$$\int_S(\vec{\nabla}\times\vec{B})\cdot d\vec{a} = \oint_P \vec{v}\cdot d\vec{l}$$

in Stoke's theorem the integral in LHS can be taken over any surface as long as it is bounded with the path P.

Therefore, $\oint\vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$ must be applicable for any surface, like ②. However, if we consider surface ② $I_{enc} = 0$: No current goes through surface ②.

Yet we have no reason to judge Ampère's law because it was derived from Biot-Savart Law.

In Maxwell's time there was no experimental reason to doubt that Ampère's law problematic. The proplem was purely theoretical and Maxwell fixed it with theoretical arguments

How Maxwell Fixed Ampère's Law

$\vec{\nabla}\cdot(\vec{\nabla}\times\vec{B}) = \mu_0(\vec{\nabla}\cdot\vec{J})$ The problematic term is $\vec{\nabla}\cdot\vec{J}$. Let's reconsider it:

$$\vec{\nabla}\cdot\vec{J} \underbrace{=}_{\text{continuity equation}} -\frac{\partial \rho}{\partial t} \underbrace{=}_{\text{Gauss Law.}} -\frac{\partial}{\partial t}(\epsilon_0 \vec{\nabla}\cdot\vec{E}) \underbrace{=}_{\text{rewrite it.}} -\vec{\nabla}\cdot\left(\epsilon_0 \frac{\partial \vec{E}}{\partial t}\right)$$

We want RHS of Ampère's law to be zero when we calculate the divergence. Therefore

$$\boxed{\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\epsilon_0\frac{\partial \vec{E}}{\partial t}}$$

In case of magnetostatics $\frac{\partial \vec{E}}{\partial t} = 0 \Rightarrow \vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

- It is hard detect $\epsilon_0 \frac{\partial \vec{E}}{\partial t}$ in the lab that's why Faraday and the others never discovered it. It was first Hertz who detected it in the Lab while he was doing some experiments with electromagnetic waves.
  Yet it plays a crucial role in the propagation of electromagnetic waves.

- Maxwell himself called this extra term displacement current:

$$\boxed{\vec{J}_d \equiv \epsilon_0 \frac{\partial \vec{E}}{\partial t}}$$

- Apart from healing the defect in Ampère's law it has a very important physical meaning:
  → Just as a changing magnetic field induces an electric field (Faraday's Law)
  it says that
  A CHANGING ELECTRIC FIELD INDUCES A MAGNETIC FIELD.

Let us see now how this extra term resolves the previous problem of a charging capacitor:

For simplicity let's consider the plates of the capacitor are very close to each other. Then the E-field between them

$$E = \frac{1}{\epsilon_0}\sigma = \frac{1}{\epsilon_0}\frac{Q}{A}$$

Q → charge on the plate

A → area of the plates.

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A}\frac{dQ}{dt} = \frac{1}{\epsilon_0 A} I$$

Therefore $\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\epsilon_0\frac{\partial \vec{E}}{\partial t}$ can be written in integral form as

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} + \mu_0\epsilon_0 \int\left(\frac{\partial \vec{E}}{\partial t}\right)\cdot d\vec{a}$$

(i) If we choose the flat surface $E = 0$ $I_{enc} = I$

(ii) If we chose surface ② $I_{enc} = 0$ but $\int\left(\frac{\partial \vec{E}}{\partial t}\right)\cdot d\vec{a} = \frac{I}{\epsilon_0}$

★ We get the same answer for both surfaces. In the first it comes from the genuine current, in the second it comes from the displacement current

Maxwell's Equations

① $\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}$ (Gauss Law)

② $\vec{\nabla}\cdot\vec{B} = 0$ (No name)

③ $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$ (Faraday's law)

④ $\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}$ (Ampère's law with Maxwell's correction)

Together with $\vec{F} = q(\vec{E}+\vec{v}\times\vec{B})$

These equations summarize the entire theoretical content of electromagnetism.

we can even derive continuity equation from them:

$$\underbrace{\vec{\nabla}\cdot(\vec{\nabla}\times\vec{B})}_{0} = \cancel{\mu_0}(\vec{\nabla}\cdot\vec{J}) + \cancel{\mu_0}\epsilon_0\frac{\partial}{\partial t}\vec{\nabla}\cdot\vec{E} \Rightarrow \vec{\nabla}\cdot\vec{J} = -\epsilon_0\frac{\partial}{\partial t}\vec{\nabla}\cdot\vec{E} = -\cancel{\epsilon_0}\frac{\partial}{\partial t}\frac{\rho}{\cancel{\epsilon_0}}$$

$$\Rightarrow \vec{\nabla}\cdot\vec{J} = -\frac{\partial}{\partial t}\vec{J}$$

If we ~~write~~ rephase this laws in the following form:

① $\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}$ ③ $\vec{\nabla}\times\vec{E} + \frac{\partial\vec{B}}{\partial t} = 0$

② $\vec{\nabla}\cdot\vec{B} = 0$ ④ $\vec{\nabla}\times\vec{B} - \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t} = \mu_0\vec{J}$

with fields $\vec{E}$ and $\vec{B}$ on the left and ~~~~ sources (ρ and $\vec{J}$) on the right it is easier to see all electromagnetic fields are created by charges and currents.

Maxwell's Equations in Matter

① $\vec{\nabla}\cdot\vec{D} = \rho_f$ ③ $\vec{\nabla}\times\vec{E} = -\partial\vec{B}/\partial t$

② $\vec{\nabla}\cdot\vec{B} = 0$ ④ $\vec{\nabla}\times\vec{H} = \vec{J}_f + \partial\vec{D}/\partial t$